

**SEC Financial
Statement Data Sets**

10-K and 10-Q filings

Matched filing-period dataset

- ✓ 331,388 observations
- ✓ 14,047 firms
- ✓ 904 AAER-positive observations

SEC AAER Releases

**Accounting and Auditing
Enforcement Releases**

Feature engineering

financial ratios
accruals
growth
Beneish-style indices
industry controls

Temporal split

Train 2009–2018
210,028 rows, 581 positives

Validation 2019–2020
45,437 rows, 156 positives

Test 2021–2023
75,923 rows, 167 positives

Model comparison

Logistic regression
Random forest
Extra trees
Gradient boosting
HistGradientBoosting

DeepCut-inspired neural thresholding

DeepCut-inspired
thresholding

Rare-event screening objective:
prioritize high-risk filings for review

Evaluation: ROC-AUC, PR-AUC, F1-score,
Top-k screening lift